

Empowering Non-Technical Users: Building Paper Circuit Boards with Zone-Based Circuit Stickers

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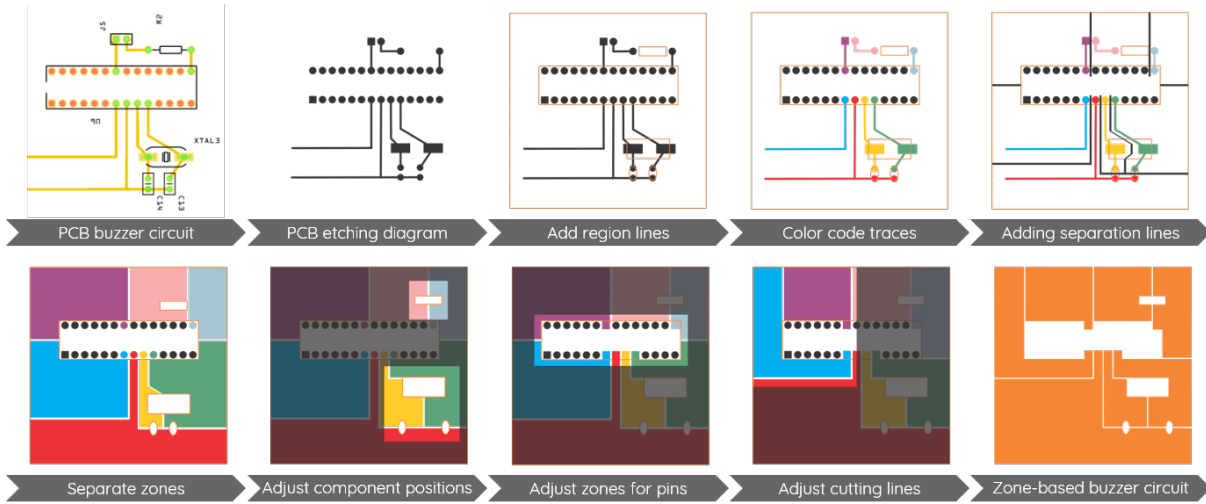


Figure 1: Steps of converting trace-based circuits to zone-based circuits

ABSTRACT

We introduce a user-friendly method for creating intricate paper circuit boards. Complex paper circuits are typically printed using conductive ink, but the expertise and experience required for this technique can be a barrier to entry for many. Our solution involves the use of zone-based circuit stickers, which empower individuals with little to no technical background to construct intricate paper circuits. We made those stickers from copper tape using a household vinyl cutter. In the presentation and demo, we will provide a detailed overview of the design process for zone-based circuits, along with insights from various design iterations. This work is an important step in democratizing the small-batch production of complex circuit boards, making them more approachable and flexible for a wider audience.

CCS CONCEPTS

• Human-centered computing → Systems and tools for interaction design.

KEYWORDS

Paper circuits, circuit stickers, fabrication, conductive zones, paper electronics.

ACM Reference Format:

Ruhan Yang and Ellen Yi-Luen Do. 2024. Empowering Non-Technical Users: Building Paper Circuit Boards with Zone-Based Circuit Stickers. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Labs (SIGGRAPH Labs '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641236.3664425>

1 INTRODUCTION

Paper circuits combine electronics and paper to provide an accessible and customizable platform for a wide range of applications [Qi and Buechley 2010; Saul et al. 2010]. The ubiquitous and flexible nature of paper makes it an ideal material for creating circuit boards that can be easily integrated into a variety of designs. Despite the great potential of paper circuit boards, there are some limitations to the current fabrication methods of paper circuits. Making delicate paper circuit boards often requires special equipment and skills [Kawahara et al. 2013], thus limiting the approachability of paper circuit board fabrication.

We propose an approachable method for creating complex and precise paper circuit boards. We begin by redesigning the circuit layout to maximize the use of conductive zones. We then created zone-based circuit stickers by subtractive processing of copper tape. These stickers enable people to rapidly build complex paper circuit boards. By enlarging the conductive zones, we ensured

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SIGGRAPH Labs '24, July 27–August 01, 2024, Denver, CO, USA

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ACM ISBN 979-8-4007-0518-2/24/07

<https://doi.org/10.1145/3641236.3664425>

greater retention of conductive material throughout the circuit, thereby increasing the efficiency of fabricating such circuits through subtractive processing. These stickers democratize the fabrication of small-scale circuit boards, requiring no special equipment or skills from the user.

2 ZONE-BASED CIRCUIT STICKERS

We propose to pre-cut copper tape and create conductive stickers for use in paper circuit board fabrication. This process uses subtractive processing. The previous methods of paper circuit fabrication primarily use additive processing, with subtractive processing receiving limited attention in this field. Subtractive processing is heavily utilized in the production of printed circuit boards (PCBs) [Coombs 2007]. The method offers several advantages over traditional additive processing, including higher precision, design flexibility, and adaptability to different materials. With this approach, we redesigned paper circuits by expanding the conductive zones. Through subtractive processing of copper tape, we created zone-based circuit stickers to help people with no technical background to make complex paper circuits.

We use a buzzer circuit as an example to experiment with paper circuit board designs. This circuit contains six components: a buzzer, a resistor, a microchip, a crystal oscillator, and two capacitors. Figure 2 shows the zone-based paper circuit and the trace-based circuit design for PCBs. We converted the circuit from trace-based to zone-based by enlarging the conductive areas.

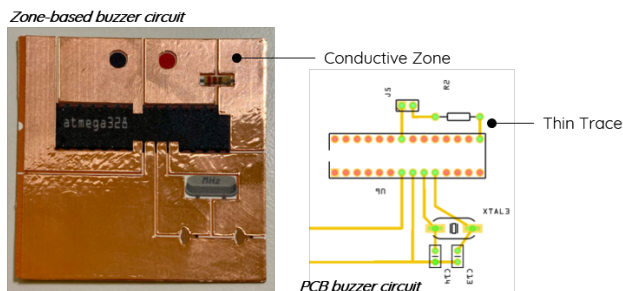


Figure 2: The zone-based buzzer circuit (left) and the trace-based buzzer circuit (right)

We converted the circuit from trace-based to zone-based by enlarging the conductive areas. Figure 1 illustrates the overview of this process, and we have divided it into 10 steps as follows:

- (1) Create a circuit design. We designed this buzzer circuit using the Fritzing [Fritzing [n. d.]].
- (2) Export pcb etching diagram. From the circuit design, we can generate the PCB etching diagram with Fritzing directly. We saved it as an editable image file.
- (3) Add region lines. We edited the etching diagram using an image editing program. First, we added region lines to the diagram. The outermost orange square is the copper tape that would be cut; the orange lines on the inside of the diagram correspond to each component that is on the circuit. This step is so that we can leave the component footprint not conductive.
- (4) Color code all traces. We marked each conductive trace with a different color to better distinguish them when we change the shapes later.
- (5) Add separation lines between the traces. We added lines between each two traces, thus dividing the entire copper tape area.
- (6) Separate conductive zones. In this step, we removed the traces and filled in the zones separated by the separation lines. These zones will fulfill the conductive function instead of the traces.
- (7) Adjust component positions. We first adjusted the position of all components, ensuring that each one is in the middle of the two zones to which it is connected. We then adjusted the position of the crystal oscillator (the rectangle on the yellow and green zones) to avoid slanted region lines. We then adjusted the position of the two capacitors (the two ovals at the bottom) to distribute them more evenly. We also adjusted the yellow and green zones along with them.
- (8) Adjust zones for pins. In this step, we added conductive zones to the microchip's pin positions.
- (9) Adjust cutting lines. We adjusted the shape of the blue and purple zones. This adjustment reduces the turns on the perimeter of the two shapes, thus reducing the difficulty of cutting during processing.
- (10) Obtain the zone-based circuit design. Lastly, we removed the unused pin markings and unified all zones to orange to represent the copper tape.

3 DEMO AND HANDS-ON EXPERIENCE

In this lab demo, we will demonstrate how to design zone-based circuit stickers and also showcase the process of making stickers using a household cutting machine. SIGGRAPH conference attendees have the opportunity to design such circuit stickers using our computers, as well as to make the stickers they designed. Participants do not need any prior knowledge of the software or how to operate the cutting machine. After the hands-on experience, participants can take away a 2-inch square piece of circuit sticker.

ACKNOWLEDGMENTS

This material is based upon work supported by the U.S. National Science Foundation under Grant No IIS-2040489.

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